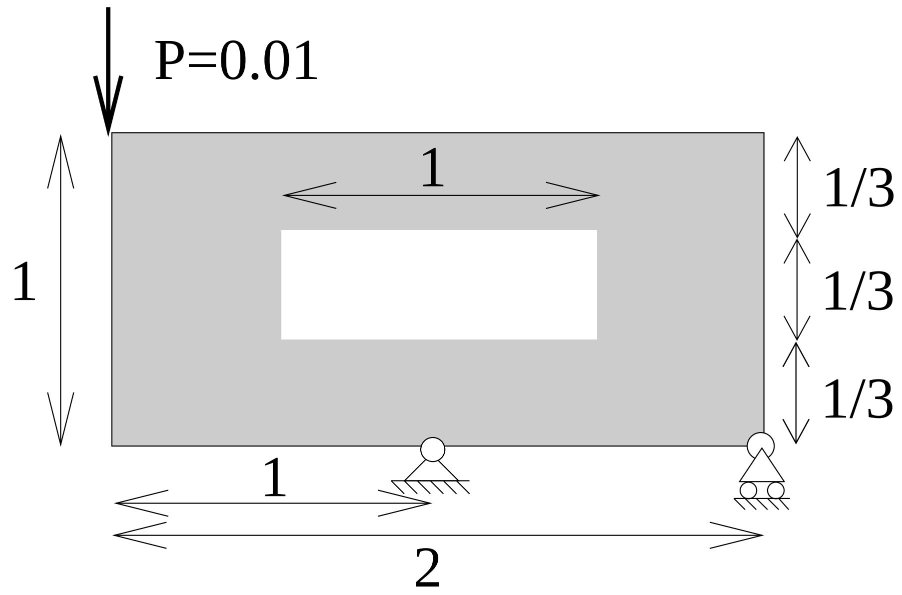
Competition, Day 1 of 41525, Finite Element Methods

Here is a competition: Who can make the stiffest truss structure? This corresponds to the truss structure with the smallest displacement (in the direction of the load) where the load is applied! [1]

Rules: The truss structure should fit into the grey domain shown below and with no bars crossing the central void area (centered white region). Position and size of the vertical load is fixed (load is applied at the upper left corner). The center of the lower edge is supported in both x and y direction and the lower right corner only in the y-direction. The bar areas and Young's moduli are fixed to $A = 1$ and $E = 1$, respectively. The maximum total length of bar elements is 10 and you may use as many nodes and elements as you want. The stiffness matrix has to be positive definite.

Create an input file called "group#.m" (where # is your group number) and upload it to Learn, September 7th (by midnight). The teachers will evaluate the proposals and announce the prize-winner in the next lecture!



[1] The submission (not the ranking!) is part of the evaluation of the first report in the course!